

Formulas for Calculating Derivatives (Homework)

Find $f'(x)$.

① $f(x) = 6\frac{1}{3}$

② $f(x) = x^7$

③ $f(x) = -9x^5$

Find $\frac{dy}{dx}$.

④ $y = 10.27$

⑤ $y = x^{51}$

⑥ $y = \frac{1}{3}x^6$

Find y' .

⑦ $y = 3\pi^2$

⑧ $y = x^{15}$

⑨ $y = -x^9$

Find $f'(x)$.

⑩ $f(x) = \frac{1}{x^3}$

$$\textcircled{11} f(x) = \frac{4}{x^{15}}$$

$$\textcircled{12} f(x) = -\frac{8}{5x^2}$$

Find $\frac{dy}{dx}$.

$$\textcircled{13} y = \sqrt{x^5}$$

$$\textcircled{14} y = \sqrt[3]{x}$$

$$\textcircled{15} y = 6\sqrt{x}$$

Find y' .

$$\textcircled{16} y = (6x)^2$$

$$\textcircled{17} y = \left(\frac{x}{5}\right)^3$$

$$\textcircled{18} y = \sqrt[4]{2x}$$

Find $f'(x)$.

$$\textcircled{19} f(x) = \sqrt{\frac{x}{10}}$$

$$\textcircled{20} f(x) = \frac{1}{\sqrt{x}}$$

$$\textcircled{21} f(x) = \frac{1}{\sqrt[4]{x^7}}$$

Find $\frac{dy}{dx}$.

$$\textcircled{22} y = (\sqrt[5]{x})^2$$

$$\textcircled{23} y = 2e^{-7}$$

$$\textcircled{24} y = x^5 \sqrt{x}$$

Find y' .

$$\textcircled{25} y = \frac{x^2}{\sqrt[4]{x}}$$

$$\textcircled{26} y = 10x$$

$$\textcircled{27} y = -\frac{2}{3}x$$

Find $f'(x)$.

$$\textcircled{28} f(x) = 8x^2 + 7x^3 - 4x + \sqrt{21}$$

$$\textcircled{29} f(x) = \frac{3}{4}x^4 + 2x^{10} - 4x^3 + 2\pi$$

$$\textcircled{30} f(x) = -0.3x^2 - x^3 + x^4$$

$$(31) f(x) = \sqrt{5x} - \left(\frac{x}{3}\right)^3 + \frac{1}{2x^9}$$

$$(32) f(x) = \sqrt[3]{x^3} + 10\sqrt[3]{x} + 8\sqrt[5]{x}$$

$$(33) f(x) = \frac{x^2}{2} - \frac{x^4}{8} + \frac{x^3}{12} + 3e^\pi$$

$$(34) f(x) = \frac{x^6}{3} + \frac{x^8}{10} - \frac{x^3}{8} + \frac{2\sqrt{7}}{5}$$

Homework Answers

$$(1) 0$$

$$(2) 7x^6$$

$$(3) -45x^4$$

$$(4) 0$$

$$(5) 51x^{50}$$

$$(6) 2x^5$$

$$(7) 0$$

$$(8) 15x^{14}$$

$$(9) -9x^8$$

$$(10) -\frac{3}{x^4}$$

$$(11) -\frac{60}{x^{16}}$$

$$(12) \frac{16}{5x^3}$$

$$(13) \frac{5}{2}\sqrt{x^3}$$

$$(14) \frac{1}{7\sqrt{x^6}}$$

$$(15) \frac{3}{\sqrt{x}}$$

$$(16) 72x$$

$$(17) \frac{3}{125}x^2$$

$$(18) \frac{\sqrt[4]{2}}{4\sqrt[4]{x^3}}$$

$$(19) \frac{1}{2\sqrt{10x}}$$

$$(20) -\frac{1}{2\sqrt{x^3}}$$

$$(21) -\frac{7}{4\sqrt[4]{x^{11}}}$$

$$(22) \frac{2}{5\sqrt[5]{x^3}}$$

$$(23) 0$$

$$(24) \frac{11\sqrt{x^9}}{2}$$

$$(25) \frac{7\sqrt[4]{x^3}}{4}$$

$$(26) 10$$

$$(27) -\frac{2}{3}$$

$$(28) 16x + 21x^2 - 4$$

$$(29) 3x^3 + 20x^9 - 12x^2$$

$$(30) -0.6x - 3x^2 + 4x^3$$

$$(31) \frac{\sqrt{5}}{2\sqrt{x}} - \frac{x^2}{9} - \frac{9}{2x^{10}}$$

$$(32) -\frac{9}{7\sqrt[7]{x^{10}}} + \frac{8}{5\sqrt[5]{x^4}}$$

$$(33) x - \frac{x^3}{2} + \frac{x^2}{4}$$

$$(34) 2x^5 + \frac{4x^7}{5} - \frac{3x^2}{8}$$